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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	newbies
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth
Project / Solution Name	Veracity
Name (Leader)	KOH HUI YUAN
Email & Phone (Leader)	huiyuankoh0409@gmail.com / +601151306389
University Affiliation(s)	Tunku Abdul Rahman University of Management and Technology (TARUMT)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small businesses—including retail shops, mini-markets, and F&B outlets—frequently manage inventory using a disjointed mix of spreadsheets, point-of-sale (POS) exports, paper delivery records, and manual adjustments. When a periodic physical count differs from the recorded system quantity (such as 43 units counted versus 50 recorded), employees face a slow and fragmented investigation process.

Staff must manually cross-check multiple disconnected sources to determine potential causes before escalating the mismatch to a manager for instructions. This manual reconciliation creates several severe operational bottlenecks:

- **Time Wasted:** Employees and managers spend valuable time searching messages and paper logs rather than focusing on core business operations.
- **Loss of Record Confidence:** Inconsistent or unsupported inventory adjustments make stock records unreliable, leading to poor reordering decisions and unverified write-offs.
- **Accountability & Traceability Gaps:** Stock numbers are frequently altered without documented reasoning or explicit manager approval, leaving owners unable to reconstruct why a correction occurred or whether it was justified.

Without a structured, verifiable reconciliation workflow, small businesses struggle to maintain inventory accuracy and financial control. **Veracity** is designed specifically for inventory staff, store managers, and SME owners to transform this recurring friction into a documented, explainable, and human-controlled review process.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Veracity helps SMEs investigate differences between recorded inventory and verified physical stock before any correction is made. It combines photo-assisted object counting, RFID-ready crate-location tracking, deterministic discrepancy assessment and manager approval.

For photo-assisted counting, staff choose a product, upload a JPG or PNG photo, and draw a box around one example item. The app returns suggested detections and a count; staff can remove false detections, manually correct missed items and confirm that the photo covers the counting area before using the result. By default, the prototype uses built-in multi-scale template matching. When optional RunPod GPU settings are configured, it calls a RunPod Serverless worker using the pre-trained OWLv2 object-detection model with the product name and example box. This is object counting, not OCR: the application does not extract text from documents or labels. AI output is only a suggestion and never changes stock automatically.

The crate module records one tag per crate, its product, declared registration quantity, recorded location and transfer history. It demonstrates tag lookup through simulated input or an entered ID from a compatible keyboard-output reader. A scan can flag a stated location that differs from the record, but does not prove physical location, transfer stock or update quantities. Physical RFID hardware integration is future work.

Confirmed counts enter a deterministic assessment engine. It compares recorded stock plus eligible unposted movements with the physical count, shows supporting, incomplete or conflicting evidence, and requires manager approval for any correction. Saved assessment versions and Decision Passports preserve the rationale and outcome.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Veracity uses a React/TypeScript and Tailwind CSS web interface, an Express/TypeScript API gateway, a Python backend and SQLite database. The frontend supports count submission, photo-count review, crate tracking, evidence review, decisions and history. Express exposes the photo-count route and forwards authenticated application requests to the Python service. The Python backend performs inventory reconciliation, permissions, crate registration/scan/transfer handling, audit logging and decision persistence in SQLite.

The default photo-count path runs locally using `sharp` and normalized template matching. If `RUNPOD_API_KEY` and `RUNPOD_ENDPOINT_ID` are configured, the same route sends the selected image, exemplar box and product label to a RunPod Serverless GPU worker running pre-trained OWLv2. The frontend receives suggested boxes, but staff must review and confirm the result. Uploaded evidence is separate from photo counting and does not itself establish a quantity.

React / TypeScript web app

├─ Count from photo → Express /api/vision/count

│ ├─ Built-in template matching (default), or

│ └─ RunPod Serverless GPU → OWLv2 detector (optional)

└─ Counts, crates, evidence and decisions → Express → Python backend → SQLite

The prototype uses no GenAI chat/completions API. Replit is a supported hosting target, while RunPod Serverless is optional GPU compute.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We designed Veracity for non-technical SME operators who need a clear explanation rather than a black-box adjustment. The workflow separates count capture, calculation and evidence review, manager approval, and decision-history replay.

The interface makes uncertainty visible. Photo detections are displayed for staff review and can be corrected before submission. RFID-related reads are explicitly labelled simulated or entered-ID input; a location mismatch prompts review rather than a stock change. The reconciliation engine exposes its calculation and distinguishes supported, incomplete and conflicting evidence.

The system can run locally through the provided launcher or be deployed on Replit. RunPod is optional; without it, the application falls back to built-in photo template matching and staff can always enter a verified count manually. Phase 2 work should validate the workflow with SME users, evaluate compatible RFID hardware, add production authentication, and measure task completion time and correction accuracy.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary SDG Target 8.2 "Improve economic productivity through technological upgrading and innovation". Secondary SDG Target 9.3**:** "Increase the access of small-scale industrial and other enterprises, in particular in developing countries, to financial services, including affordable credit, and their integration into value chains and markets."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	45 minutes median investigation time per discrepancy with an estimated 60% untraceable adjustment rate (based on retail SME operational benchmarks, staff spend ~45 minutes manually cross-checking paper logs, WhatsApp messages, and spreadsheets per mismatch, with most adjustments applied without complete audit trails). *(Source: World Bank MSME Digitalization Benchmarks / Retail Operations Loss Prevention Manuals).*
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Median minutes per discrepancy from count submission to documented decision (measured across standardized test scenarios) and the share of completed cases with a saved Decision Passport and supporting evidence records.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	80% reduction in median resolution time (reducing discrepancy processing from 45 minutes to under 9 minutes per incident) and achieving a 100% auditability rate (100% of completed stock adjustments backed by a

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	saved Decision Passport with immutable supporting evidence).

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type	Licence / use
React, TypeScript, Vite, Tailwind CSS, Express, Sharp, SQLite and other project dependencies	Open-source libraries/frameworks	Used under respective upstream licences
OWLv2 (`google/owlv2-base-patch16-ensemble`)	Pre-trained object-detection model	Optional RunPod GPU photo-count worker; not trained or fine-tuned by the team

PyTorch, Transformers, NumPy, OpenCV, Pillow, SciPy, RunPod SDK	Open-source libraries/SDKs	Used by the optional OWLv2 worker
Project CSVs and photo fixtures	Synthetic demo/evaluation data	No personal or sensitive data intended

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



KOH HUI YUAN
Team leader - newbies

24 October 2026